Bayes Theorem

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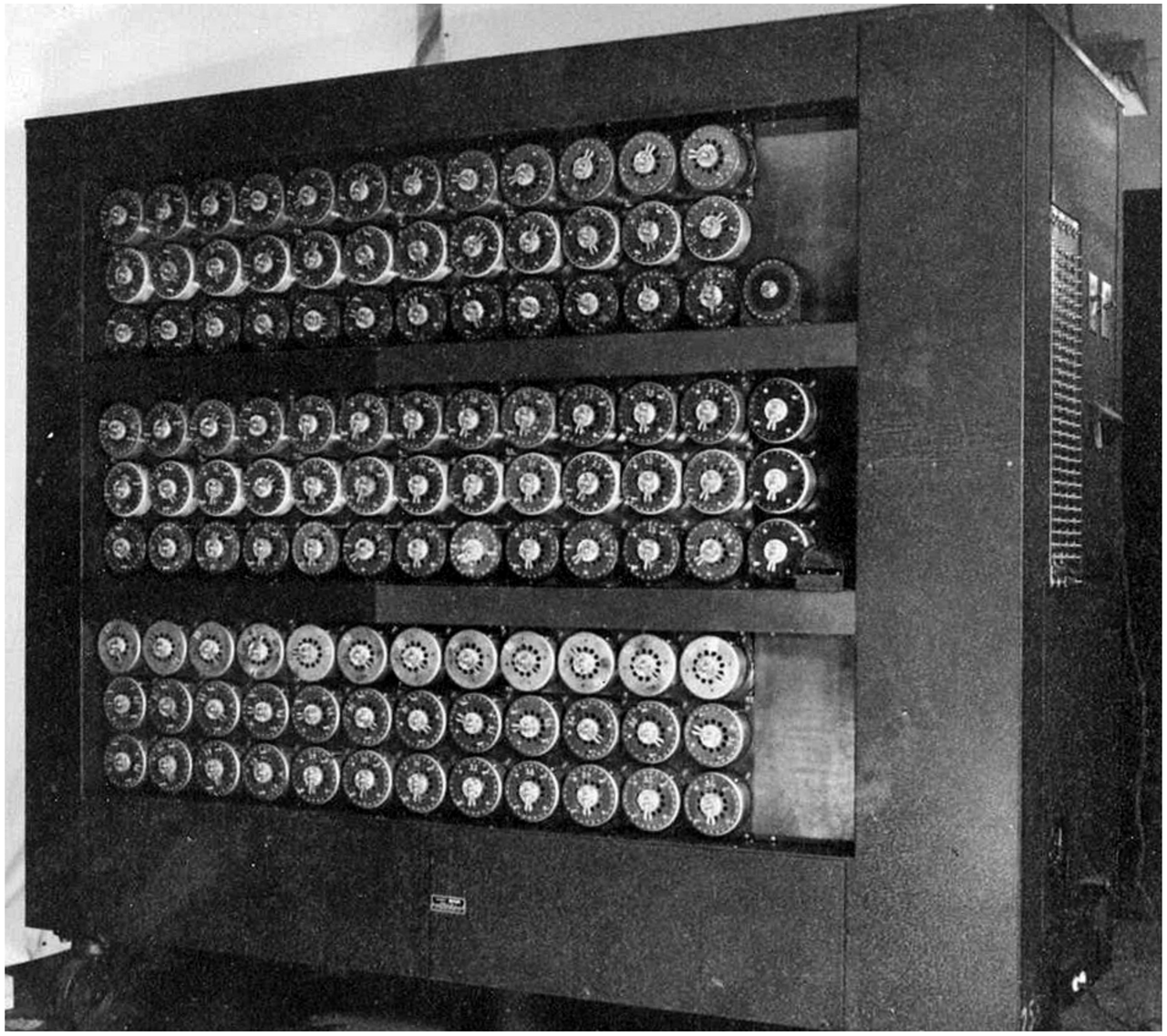
Credit to:

- Tom Hobbs (Colorado State University)
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What is the collective noun/term for a group of

- statisticians?
- Ecologists?





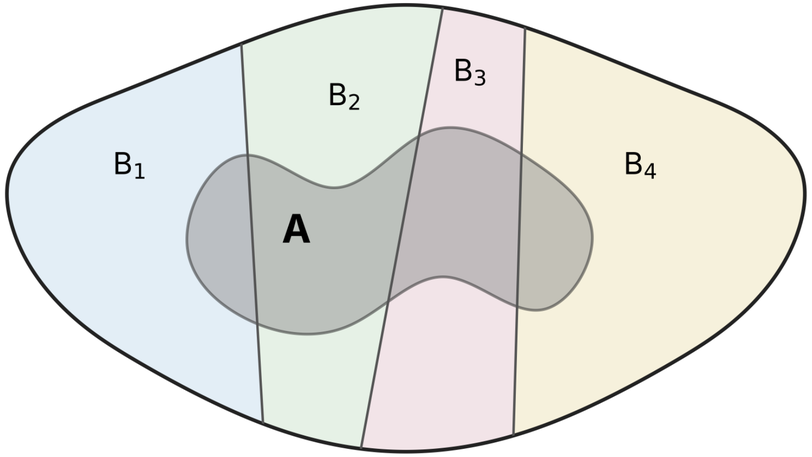
Wikipedia

Bayes Theorem

- All unobserved quantities are Random Variables

- Bayes Theorem:

$$[\theta|y] = \frac{[y|\theta]\,[\theta]}{[y]}$$



- $[A] = \sum_i [A \mid B_i]\,[B_i]$ - From Law of Total Probability
- $[A] = \int_B [A|Bi][B]dB$ - For continuous Random Variables
  - Simialrly $[y] = \int [y|\theta]\,[\theta]d\theta$ for continuous parameters

- Thus, Bayes Theorem:

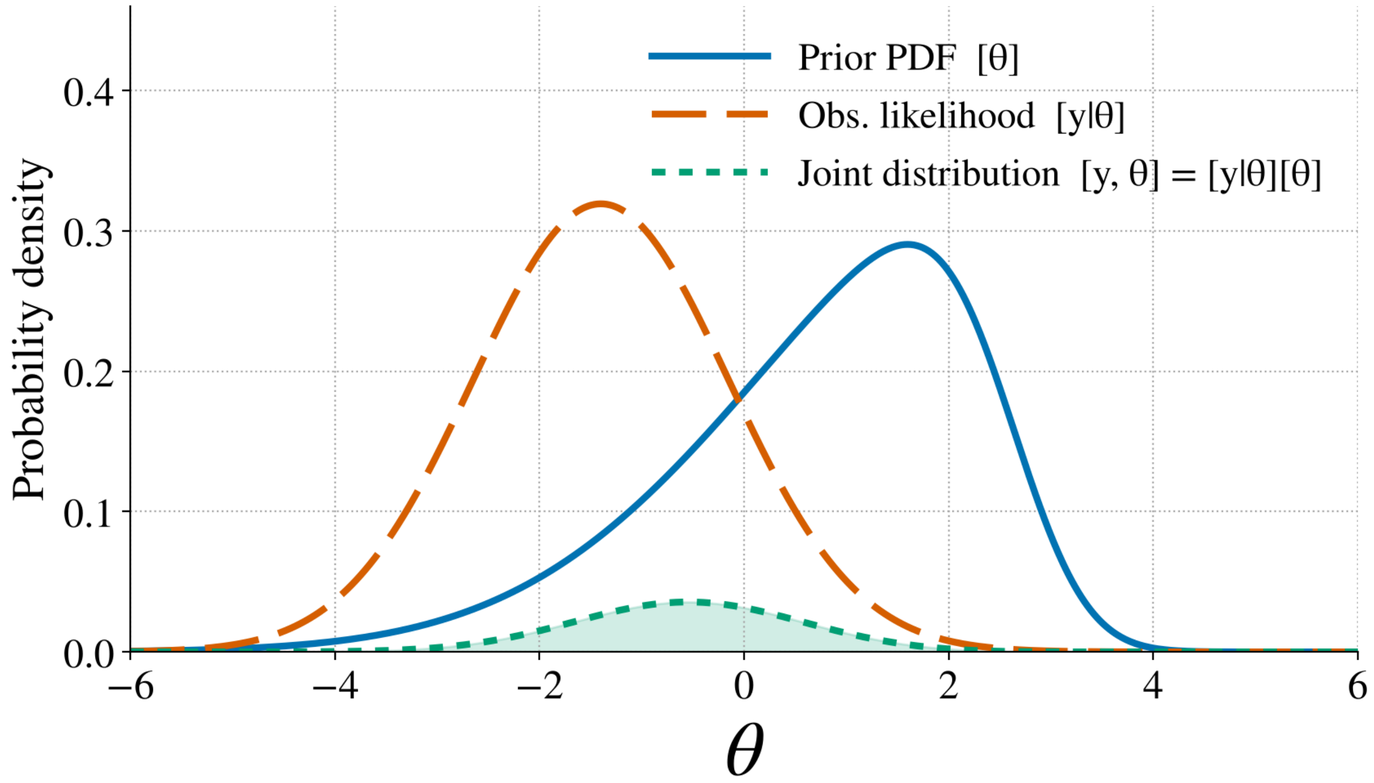
$$[\theta|y] = \frac{[y|\theta]\,[\theta]}{\int [y|\theta]\,[\theta]d\theta}$$

Bayes Theorem

- $[y] = \int [y|\theta]\,[\theta]d\theta$ is a constant after data is observed
  - It is also known as marginal distributions
- $[\theta|y] \propto [y|\theta]\,[\theta]$
  - $[\theta|y]$ - Posterior
  - $[y|\theta]$ - Liklihood
  - $[\theta]$ - Prior
  - $[y|\theta]\,[\theta]$ - Joint Distribution

Component of Bayes Theorem

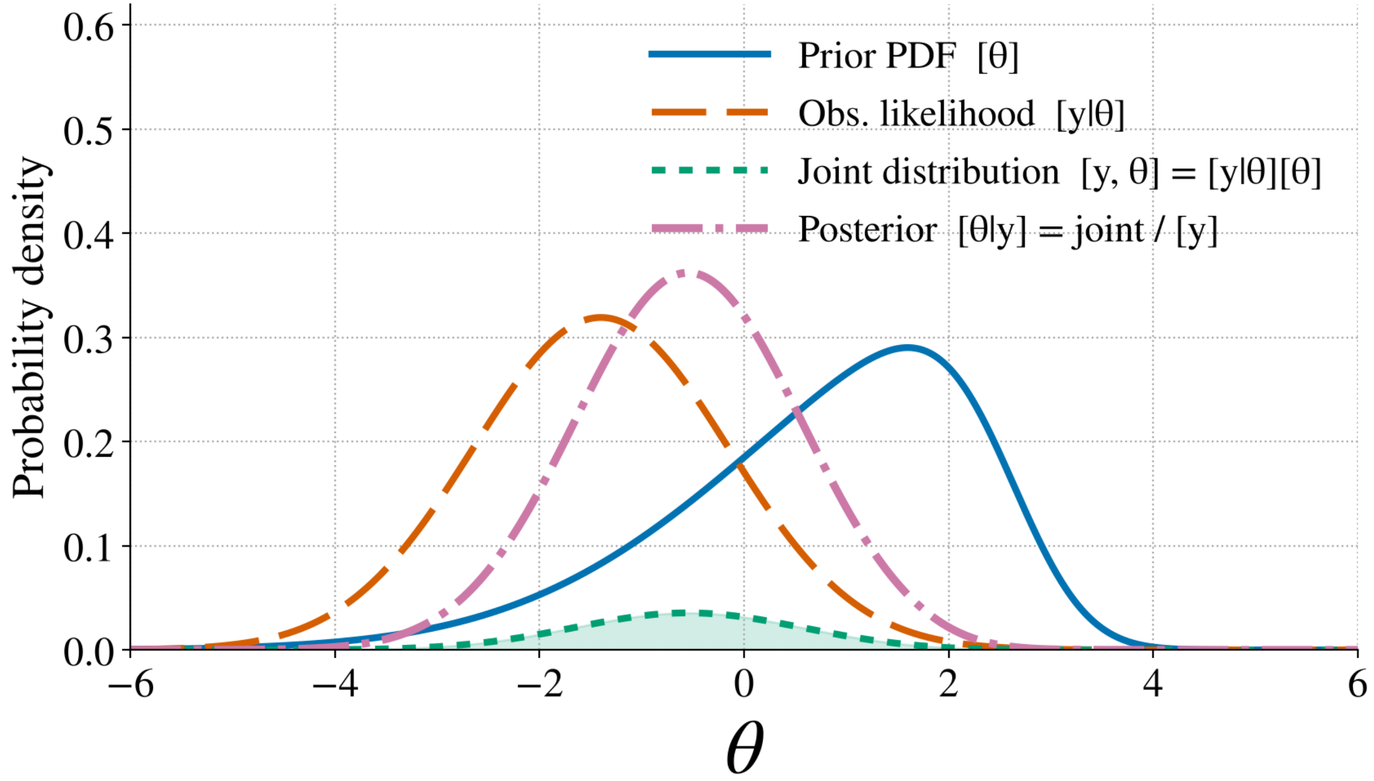
$$[\theta \mid y] = \frac{[y, \theta]}{[y]} = \frac{[y \mid \theta]\,[\theta]}{\int_\theta [y \mid \theta]\,[\theta]\,d\theta}$$



Shaded region - Marginal Distribution, it is the area under joint distribution

Component of Bayes Theorem

$$[\theta \mid y] = \frac{[y, \theta]}{[y]} = \frac{[y \mid \theta]\,[\theta]}{\int_\theta [y \mid \theta]\,[\theta]\,d\theta}$$

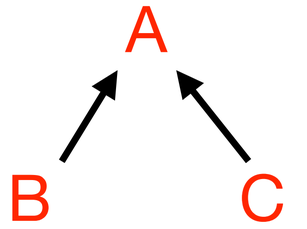


Shaded region - Marginal Distribution, it is the area under joint distribution

Convert this Theory to Application: Directed Acyclic Graphs Joint Distributions

Directed Acyclic Graphs: DAGs

- DAGs - Bayesian Network
  - How Joint distributions are factored into conditional distributions using nodes to represent Random Variables and arrows represent conditioning.
- Nodes at the heads of arrow should be on the left hand side of the conditioning symbol: “|”
- Nodes at the tail of arrow without an arrow leading to it must be expressed unconditionally.



[A|B,C] [B] [C]